

Part I

SUMMARY of Understanding Functions

Summary

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What we need to know about functions

- State the definition of a function.
- Find the domain and range of a function.
- Distinguish between functions by considering their domains.
- Recognize different representations of the same function.
- Determine where a function is positive or negative.
- Determine algebraically whether a function is even, odd, or neither.
- Use symmetry when graphing even or odd functions.
- Plot basic functions.
- Apply appropriate transformations to graphs of basic functions (vertical and horizontal shifts, vertical and horizontal stretching and reflecting)
- Perform basic operations and compositions on functions.
- Work with piecewise defined functions.
- Determine if a function is one-to-one.
- Define and work with inverse functions.
- Plot inverses of basic functions.
- Find inverse functions (algebraically and graphically).
- Find the largest interval containing a given point where the function is invertible.
- Determine the intervals on which a function has an inverse.
- Relate the domain and range of f and the domain and range of f^{-1} .
- Use a sign-chart to determine where the function is positive/negative.

Part II

Recitation Questions

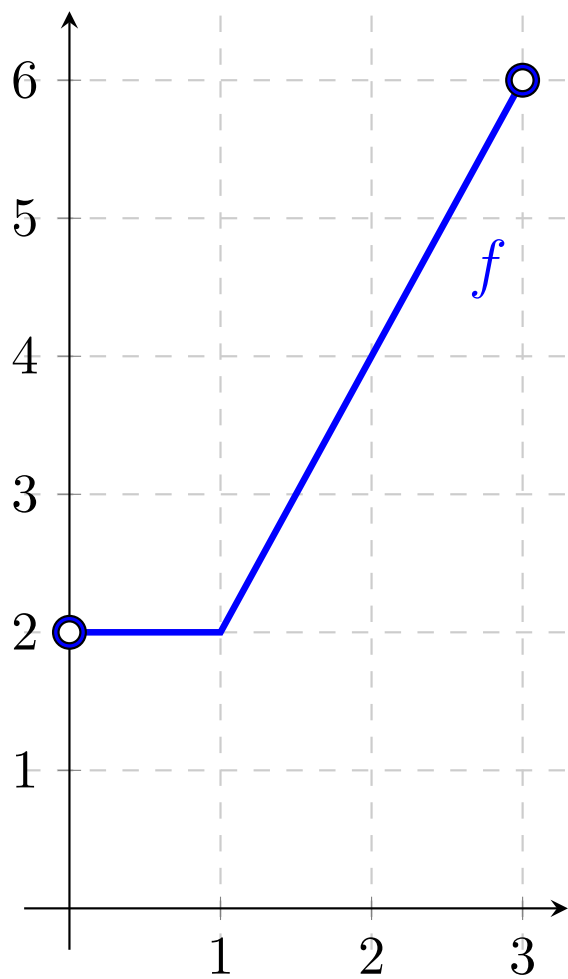
Problem 1

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Problem 1

Problem 1 Find a formula for the function f whose graph is given in the figure below.

What is the domain of f ? What is the range of f ? Is the function f linear?

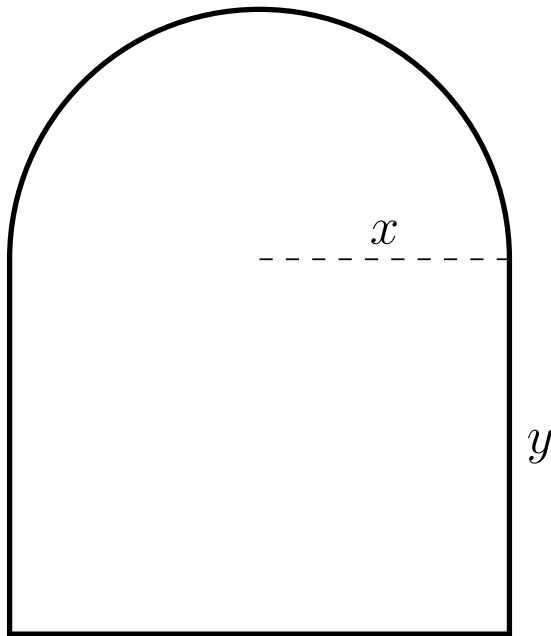


Problem 2

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Problem 2

Problem 2 A mirror has the shape of a rectangle surmounted by a semicircle (see figure). The area of the mirror is 32 in^2 . Let x be the radius of the semicircle (that lies on top of the rectangle). Express the perimeter of the mirror P (in inches) as a function of x (in inches). Is P a polynomial, a rational function or a transcendental function?



Problem 3

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Problem 3

Problem 3 Define $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 0 \\ ? & \text{if } x > 0 \end{cases}$

- (a) Find an expression for "?" such that f will be even.
- (b) Find an expression for "?" such that f will be odd.

Problem 4

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Problem 4

Problem 4 Given $y(t) = t - \frac{\pi}{3}$ and $w(t) = \sin(t)$. Find:

(a) $y(w(t))$

(b) $w(y(t))$

(c) $w\left(y\left(\frac{4\pi}{3}\right)\right)$

(d) $y(w(\frac{4\pi}{3}))$

Problem 5

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Problem 5

Problem 5 Define $g(x) = \begin{cases} |x - 2| & \text{if } x < 0 \\ \cos(x) & \text{if } x \geq 0 \end{cases}$

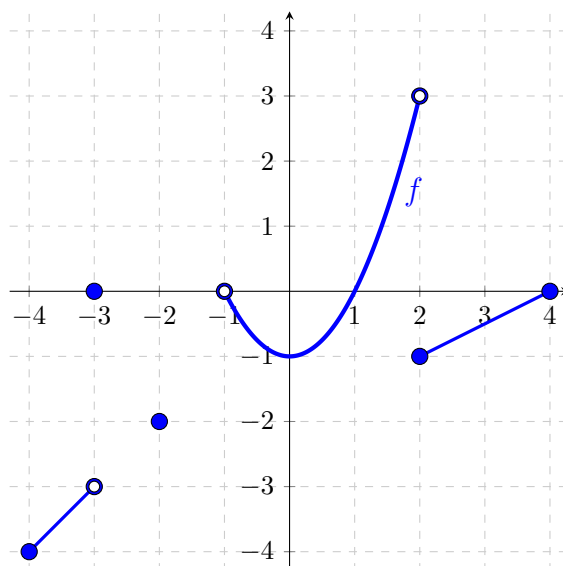
- (a) *Sketch a graph of g*
- (b) *Find the domain and range of g*
- (c) *Find the values of $g(\pi)$ and $g(-\pi)$*

Problem 6

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Problem 6

Problem 6 The entire graph of $f(x)$ is given below.



- Find the domain and range of f
- Find the values of $f(-3)$, $f(-2)$, $f(-1)$, $f(2)$
- Find the intervals on which $f(x)$ is positive. Find the intervals on which $f(x)$ is negative.
- Find the intervals on which f is increasing. Find the intervals on which f is decreasing.
- True or False: $f(1.5) < f(2)$

Problem 7

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Problem 7

Problem 7 *Determine if the function is even, odd, or neither.*

(a) $h(x) = x^4 + x^2 - 3$

(b) $s(t) = t^2 - t$

(c) *We know that $\sin(\theta)$ is odd and $\cos(\theta)$ is even. Is $g(\theta) = \tan(\theta)$ even, odd, or neither?*

Problem 8

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Problem 8

Problem 8 Using the known graphs of $y = \sqrt{x}$ and $y = \frac{1}{x}$, sketch the graphs of the following using transformations.

(a) $y = \sqrt{x+2} - 3$

(b) $y = \frac{1}{x-4} + 1$

Problem 9

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Problem 9

Problem 9 Find the domain of the function. Determine whether the function is odd, even, or neither.

(a) $f(x) = \frac{x}{\sqrt{x^2 - 9}}$

(b) $g(x) = \frac{\sin(x)}{x}$

(c) $h(t) = \ln(t^3 - 1)$

Problem 10

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Problem 10

Problem 10 Let g be a one-to-one function and let g^{-1} be its inverse. **True or False:** If the point $(2, 1/5)$ lies on the graph of g , then the point $(2, 5)$ lies on the graph of g^{-1} .

Problem 11

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Problem 11

Problem 11 Each of the following functions are invertible on their given domains. For each one find a formula for its inverse and give the domain and range of the inverse.

- (a) The function f defined by $f(x) = x^2 - 4x - 5$ for every $x \geq 2$.
- (b) The function g defined by $g(u) = \sqrt[4]{u + 2}$.
- (c) The function h defined by $h(t) = 1/(t + 2)^2$ for every $t > -2$.

Problem 12

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Problem 12

Problem 12 *Explain what each of the following means:*

(a) $f^{-1}(x)$

(b) $f(x^{-1})$

(c) $(f(x))^{-1}$

Problem 13

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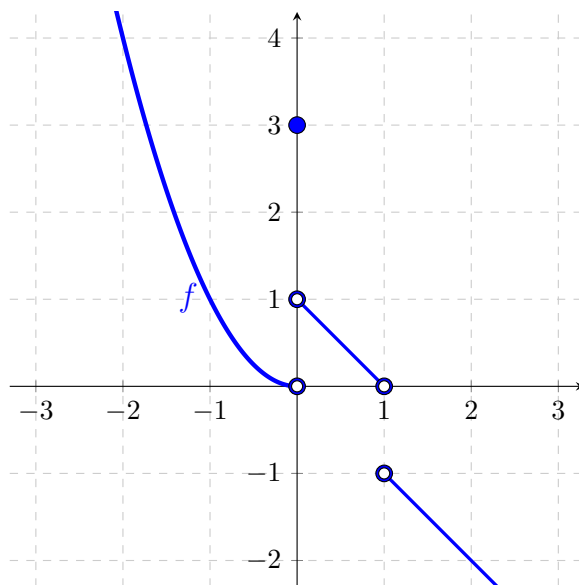
Problem 13

Problem 13 *If $f(x)$ represents the number of packages of buns needed for x packages of hotdogs, what does $f^{-1}(x)$ represent?*

Problem 14

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Problem 14 We're given the following graph of a function:



Use this graph to answer the following questions:

- What is the domain of this function?
- What is the range of this function?
- What is the value of $f(0)$, $f(1)$, and $f(2)$?
- Does this function have an inverse? (Why or why not?)
- Find at least two intervals on which the function is one-to-one.
- Find $f^{-1}(3)$ on a restricted domain of f .
- Using the restricted domain for f of $(-\infty, 0)$, sketch a graph of f^{-1} .
- Using the restricted domain for f of $[0, 1) \cup (1, \infty)$, sketch a graph of f^{-1} .